APPROXIMATION BY PERFECT COMPLEXES OVER AFFINE SPACES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-perfect.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof descends E along epsilon, transfers pseudo-coherence and support by the stalk comparison, applies the scheme approximation lemma, and pulls the perfect approximation back using descent of perfectness

Editorial note. The label lemma-approximation-affine names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- the definition of approximation for triples (T, E, m) (definition-approximation-holds of the same chapter)
- the comparison of D_QCoh on the small etale site with the Zariski site via epsilon (lemma-derived-quasi-coherent-small-etale-site of the same chapter)
- descent of m-pseudo-coherence along epsilon (lemma-descend-pseudo-coherent of the same chapter) and the stalk comparison of cohomology sheaves (as in the proof of lemma-epsilon-flat)
- the affine scheme approximation lemma (Derived Categories of Schemes, Lemma perfect-lemma-approximation-affine)
- descent of perfectness along epsilon (lemma-descend-perfect of the same chapter)

Source reference. Original reference 08EM: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 3575–3575. Prerequisite pointer invoked at proof line(s) 3654; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original equation 08H7: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 320–320. Prerequisite pointer invoked at proof line(s) 3668; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 071Q: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 402–402. Prerequisite pointer invoked at proof line(s) 3670; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:spaces-perfect.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 08HG: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 3288–3288. Prerequisite pointer invoked at proof line(s) 3678; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 08HE: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 3133–3133. Prerequisite pointer invoked at proof line(s) 3672; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 08H8: sources/stacks-spaces-perfect/pinned-original-source.tex, source lines 330–330. Prerequisite pointer invoked at proof line(s) 3674; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 08EQ: sources/stacks-chow/tags.txt, source lines 10681–10681. Prerequisite pointer invoked at proof line(s) 3676; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let S be a scheme. Let X be an algebraic space over S which is representable by an affine scheme. Then approximation holds for every triple (T, E, m) as in Definition 08EM such that there exists an integer $r \geq 0$ with*

(1) *E is m-pseudo-coherent,*
(2) *$H^i(E)$ is supported on T for $i \geq m - r + 1$,*
(3) *$X \setminus T$ is the union of r affine opens.*

In particular, approximation by perfect complexes holds for affine schemes.

Proof. Let X_0 be an affine scheme representing X. Let $T_0 \subset X_0$ by the closed subset corresponding to T. Let $\epsilon : X_{étale} \to X_{0,Zar}$ be the morphism (08H7). We may write $E = \epsilon^* E_0$ for some object E_0 of $D_{QCoh}(\mathcal{O}_{X_0})$, see Lemma 071Q. Then E_0 is m-pseudo-coherent, see Lemma 08HE. Comparing stalks of cohomology sheaves (see proof of Lemma 08H8) we see that $H^i(E_0)$ is supported on T_0 for $i \geq m-r+1$. By Derived Categories of Schemes, Lemma 08EQ there exists an approximation $P_0 \to E_0$ of (T_0, E_0, m). By Lemma 08HG we see that $P = \epsilon^* P_0$ is a perfect object of $D(\mathcal{O}_X)$. Pulling back we obtain an approximation $P = \epsilon^* P_0 \to \epsilon^* E_0 = E$ as desired. □